Pipes

meghana@hp1: ~

GNU nano 8.7.1 pipe.c *

```c
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd[2];
    char message[] = "Hello from Parent";
    char buffer[100];

    pipe(fd);

    if (fork() == 0)
    {
        // Child process
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Received: %s\n", buffer);

        close(fd[0]);
    }
    else
    {
        // Parent process
        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        printf("Message sent\n");

        close(fd[1]);
    }

    return 0;
}
```

```
meghana@hp1:~$ nano pipe.c
meghana@hp1:~$ gcc pipe.c -o pipe
meghana@hp1:~$ ./pipe
message sent
Received: Hello from Parent
meghana@hp1:~$
```

Pipe reader

meghana@hp1: ~

```
 GNU nano 8.7.1                          pipe_reader.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    int fd[2];
    char buffer[100];

    pipe(fd);

    if (fork() == 0)
    {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Received: %s\n", buffer);

        close(fd[0]);
    }
    else
    {
        close(fd[0]);

        write(fd[1], "Hello from Writer", 18);

        close(fd[1]);
    }

    return 0;
}
```

```
meghana@hp1:~$ nano pipe_reader.c
meghana@hp1:~$ gcc pipe_reader.c -o pipe_reader
meghana@hp1:~$ ./pipe_reader
Received: Hello from Writer
meghana@hp1:~$
```